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ACH File Upload API

Integration Guide

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1. Introduction

BOC Bank provides a secure RESTful API for submitting ACH debit files in standard NACHA format. This guide covers everything you need to integrate with our ACH File Upload service, including authentication, file requirements, the upload process, response handling, and error resolution.

The API accepts a complete NACHA file, validates it against your organization's configuration, and, if valid, queues every entry for processing through BOC Bank's ACH pipeline. The entire file is either accepted or rejected; there is no partial processing.

Who should read this guide

This guide is intended for developers and technical staff responsible for integrating your application with BOC Bank's ACH origination service. Familiarity with the NACHA file format and RESTful APIs is assumed.

2. Getting Started

Prerequisites

Before you begin, ensure the following are in place:

- Your organization has been onboarded with BOC Bank and has an active ACH origination agreement.
- You have received your API key from BOC Bank. Keys follow the format `boc_XXXXXXXX_...` and should be stored securely.
- Your server's public IP address has been registered with BOC Bank for network allowlisting (if applicable).
- You can generate valid NACHA-formatted ACH files from your application.

Base URL

Your BOC Bank representative will provide the development and production base URLs during onboarding. All API calls use HTTPS (TLS 1.2 or higher is required).

Authentication

Every request must include your API key in the X-API-Key header. The API key is the sole authentication mechanism for this endpoint. No additional tokens, sessions, or OAuth flows are required.

```
X-API-Key: boc_XXXXXXXX_your_secret_key_here
```

Your key is tied to your organization and controls which ACH configuration (Company ID, transaction types, daily limits) applies to your uploads. If your key is compromised, contact BOC Bank immediately to have it revoked and a new key issued.

3. File Requirements

NACHA format

Files must conform to the standard NACHA ACH file format. Each record is exactly 94 characters, and the file must contain the standard record types in the correct order.

Record Type	Code	Description
File Header	1	One per file. Contains routing and format information.
Batch Header	5	One per batch. Contains your Company ID, SEC code, and effective date.
Entry Detail	6	One per transaction. Contains the receiving account, amount, and individual identification.
Addenda	7	Optional. Accepted but not extracted in current version.
Batch Control	8	One per batch. Contains entry counts, hash totals, and amount totals.
File Control	9	One per file. Contains batch counts, entry counts, and file-level totals.

Key fields validated by BOC Bank

Field	Location	Requirement
Company Identification	Batch Header (pos 41–50)	Must match the Company ID registered with BOC Bank for your organization.
SEC Code	Batch Header (pos 51–53)	Must match your authorized Standard Entry Class code (e.g., PPD).
Transaction Code	Entry Detail (pos 2–3)	Must be in your approved list of transaction codes (e.g., 27 for debit, 37 for prenote).
Routing Number	Entry Detail (pos 4–12)	Must pass mod-10 check digit validation and exist in the Federal Reserve routing directory.
Effective Date	Batch Header (pos 70–75)	Must fall within the allowed date window (typically the next banking day through two banking days ahead).
Control Totals	Batch & File Control	Entry counts, hash totals, and debit totals must balance across batch and file control records.



File size and naming

Maximum file size is configured per organization (default: 10 MB). You may optionally provide a filename via the x-file-name header. If omitted, the system will generate one. We recommend a convention such as rent_YYYYMMDD.ach for your own traceability.

4. Uploading a File

Request format

```
POST /api/v1/ach/files
```

Headers:

```
X-API-Key: boc_XXXXXXX_your_key  
Content-Type: application/octet-stream  
Content-Length: {file-size-in-bytes}  
x-file-name: rent_20260502.ach    (optional)
```

Body: Raw NACHA file content

Send the entire NACHA file as the raw request body. Do not base64-encode it, wrap it in JSON, or use multipart form encoding. The Content-Type must be application/octet-stream.

What happens when you upload

The API processes your file through a series of validation stages, in order:

Stage	What is checked	If it fails
1. Authentication	API key validity, role authorization, network allowlisting, active ACH configuration	401 or 403
2. File Size	File is not empty and does not exceed your configured maximum	400 or 413
3. Duplicate Detection	SHA-256 hash compared to previously uploaded files for your organization	409
4. NACHA Validation	File structure, record lengths, Company ID, SEC code, transaction codes, routing numbers, control totals	422
5. Business Rules	Effective date within allowed window, total debits within your daily limit	403 or 422
6. Persistence	All entries recorded and queued for bank approval and processing	200 (success)

If any stage fails, the entire file is rejected and no entries are created. The response will include a specific error code and message to help you identify and resolve the issue.

5. Understanding the Response

Successful upload (200 OK)

When your file passes all validation stages, you receive a 200 response with details about each entry:

```
{
  "fileId": "F-00123",
  "status": "Accepted",
  "receivedDate": "2026-05-01T14:30:00Z",
  "totalEntries": 3,
  "totalDebitAmount": 3750.00,
  "items": [
    {
      "bankReference": "BPY-98765",
      "traceNumber": "091000010000001",
      "individualId": "a1b2c3-...",
      "individualName": "JOHN SMITH",
      "amount": 1250.00,
      "transactionCode": "27",
      "effectiveDate": "2026-05-02",
      "status": "Accepted"
    },
    ...
  ]
}
```

Key response fields

Field	Description
fileId	Unique identifier for this upload. Use this to reference the file in future status queries.
status	File-level status. Will be "Accepted" on a successful upload.
receivedDate	UTC timestamp of when BOC Bank received the file.
totalEntries	Number of individual entries (transactions) in the file.
totalDebitAmount	Total debit amount across all entries, in dollars.
items[].bankReference	BOC Bank's internal reference for each entry. Save this for reconciliation and status lookups.
items[].traceNumber	The trace number from your NACHA file, echoed back for your reference.
items[].individualId	The Individual Identification field from your entry detail records.



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Note: "Accepted" means the file has passed validation and entries are queued for processing. It does not mean funds have been debited. Entries will go through bank approval and then be transmitted to the Federal Reserve on the effective date.

6. Entry Lifecycle

After a successful upload, each entry moves through the following stages:

Stage	What happens	Typical timing
Accepted	File passes validation; entries are queued for bank approval.	Immediate (synchronous response)
Admin Approved	A BOC Bank officer reviews and approves the entries.	Same day (business hours)
Transmitted	Entries are included in an outbound NACHA file sent to the Federal Reserve.	Same day or next business day
Settled	The return window has passed with no return received. Funds are considered settled.	T+2 business days after transmission
Returned	The receiving bank has returned the entry (e.g., insufficient funds, account closed).	T+1 to T+2 business days
NOC Applied	A Notification of Change has been received with corrected account information.	T+1 to T+2 business days

A returned entry will not settle. An NOC can be received on either a settled or unsettled entry and indicates that the receiving bank has corrected information (such as an account number) for future transactions.

Note: A Transaction Status API is planned for a future release, which will allow you to query the current status and full event history of any entry by file ID, trace number, or individual ID.

7. Error Reference

When a file is rejected, the response includes an HTTP status code and a JSON body with an error code and human-readable message. Below is a complete reference of all possible error responses.

Authentication and authorization errors

HTTP Status	Error Code	Meaning	What to do
401	MISSING_API_KEY	No X-API-Key header was provided.	Include your API key in the X-API-Key header.
401	INVALID_API_KEY	The key was not recognized, has been revoked, or is inactive.	Verify the key is correct. Contact BOC Bank if it has been revoked.
401	API_KEY_EXPIRED	The key has passed its expiration date.	Contact BOC Bank to issue a new key.
403	NETWORK_RESTRICTED	Your server's IP address is not in the allowed range for this key.	Verify you are calling from the registered IP. Contact BOC Bank to update your allowlist.
403	ACH_NOT_CONFIGURED	No active ACH configuration exists for your organization.	Contact BOC Bank to complete your ACH setup.

File-level errors

HTTP Status	Error Code	Meaning	What to do
400	EMPTY_BODY	The request body was empty.	Ensure the NACHA file content is included in the request body.
409	DUPLICATE_FILE	This exact file has already been uploaded.	The response includes the existing file ID. Do not resubmit the same file.
413	FILE_TOO_LARGE	The file exceeds your configured maximum size.	Split the file into smaller batches or contact BOC Bank to increase your limit.
429	RATE_LIMIT_EXCEEDED	Too many requests in a short period.	Wait for the duration indicated in the Retry-After header before retrying.

NACHA validation errors

HTTP Status	Error Code	Meaning
422	INVALID_RECORD_LENGTH	One or more records is not exactly 94 characters.
422	INVALID_FILE_HEADER	The file header record is malformed or contains incorrect routing information.
422	MISSING_FILE_CONTROL	The file does not end with a valid file control record.
422	BATCH_STRUCTURE_ERROR	Batches are not properly opened (Type 5) and closed (Type 8).
422	COMPANY_ID_MISMATCH	The Company Identification in the batch header does not match your registered Company ID.
422	SEC_CODE_MISMATCH	The Standard Entry Class code does not match your authorized SEC code.
422	INVALID_TRANSACTION_CODE	A transaction code in an entry is not in your approved list.
422	INVALID_ROUTING_CHECK_DIGIT	The check digit on a receiving bank routing number failed mod-10 validation.
422	INVALID_ROUTING_NUMBER	A receiving bank routing number was not found in the Federal Reserve directory.
422	INVALID_PRENOTE_AMOUNT	A prenote entry (transaction code 37) has a non-zero amount.
422	INVALID_EFFECTIVE_DATE	The effective date is outside the allowed window (too far in the past or future).
422	BATCH_ENTRY_COUNT_MISMATCH	The entry count in a batch control record does not match the actual number of entries.
422	BATCH_HASH_MISMATCH	The entry hash in a batch control record does not match the calculated hash.
422	BATCH_DEBIT_TOTAL_MISMATCH	The total debits in a batch control record do not match the sum of entry amounts.
422	FILE_BATCH_COUNT_MISMATCH	The batch count in the file control does not match the actual number of batches.
422	FILE_ENTRY_COUNT_MISMATCH	The entry count in the file control does not match the sum of batch entry counts.
422	FILE_HASH_MISMATCH	The entry hash in the file control does not match the sum of batch hashes.
422	FILE_TOTAL_MISMATCH	The total debits in the file control do not match the sum of batch totals.
422	EMPTY_FILE	The file contains no entry detail records.

Business rule errors

HTTP Status	Error Code	Meaning	What to do
403	EXPOSURE_LIMIT_EXCEEDED	The total debits in this file, combined with other files uploaded today, would exceed your daily debit limit.	Wait until the next calendar day, or contact BOC Bank to adjust your limit if needed.

8. Best Practices

File preparation

- Validate your NACHA files locally before uploading. Ensure record lengths are exactly 94 characters and control totals balance.
- Use a consistent file naming convention (e.g., rent_YYYYMMDD.ach) and pass it in the x-file-name header for traceability.
- Verify that your Company ID, SEC code, and transaction codes match what BOC Bank has configured for your organization.

Error handling

- Always check the HTTP status code first. A 200 response means the file was accepted; any other code means the entire file was rejected.
- Parse the error code from the JSON response body to determine the specific issue. The error message provides additional detail.
- For 409 (duplicate) responses, do not retry. The response includes the original file ID for your reference.
- For 429 (rate limit) responses, wait for the duration specified in the Retry-After response header before retrying.

Security

- Store your API key securely (e.g., in a vault or secrets manager). Never embed it in client-side code or commit it to source control.
- Use HTTPS for all API calls. The API enforces TLS 1.2 or higher.
- Rotate your API key periodically. BOC Bank supports multiple active keys per organization, so you can provision a new key before revoking the old one.
- Restrict API calls to known server IP addresses. BOC Bank can configure per-key network allowlisting for additional security.

Reconciliation

- Save the bankReference returned for each entry in your successful upload response. This is the primary identifier for status lookups and reconciliation.
- Save the fileId for file-level tracking and duplicate reference.



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- The traceNumber echoed in the response matches what you submitted in your NACHA file and can be used to correlate entries between systems.

9. Limits and Configuration

The following limits are configured per organization by BOC Bank. Contact your BOC Bank representative if you need adjustments.

Parameter	Description	Default
Maximum file size	Largest NACHA file accepted in a single upload	10 MB
Daily debit limit	Maximum total debits across all files uploaded in a calendar day	Set during onboarding
Effective date window	How far in advance the effective date can be from the next banking day	2 banking days
Allowed transaction codes	Which NACHA transaction codes your files may contain	Set during onboarding
Same-day ACH	Whether same-day effective dates are permitted	Disabled by default
Rate limit	Maximum number of upload requests per minute	10 per minute

10. Support

If you encounter issues during integration or production use, contact BOC Bank through your designated relationship manager or the channels provided during onboarding.

When reporting an issue, please include:

- The full HTTP status code and error code from the response.
- The fileId (if the file was previously accepted) or the x-file-name you submitted.
- The approximate date and time (UTC) of the request.
- Any relevant details about the file contents (number of entries, total amount, effective date).

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